

**Supplementary Information for Smart eDNA sampling strategies: Spatial autocorrelation
can guide efficient survey design**

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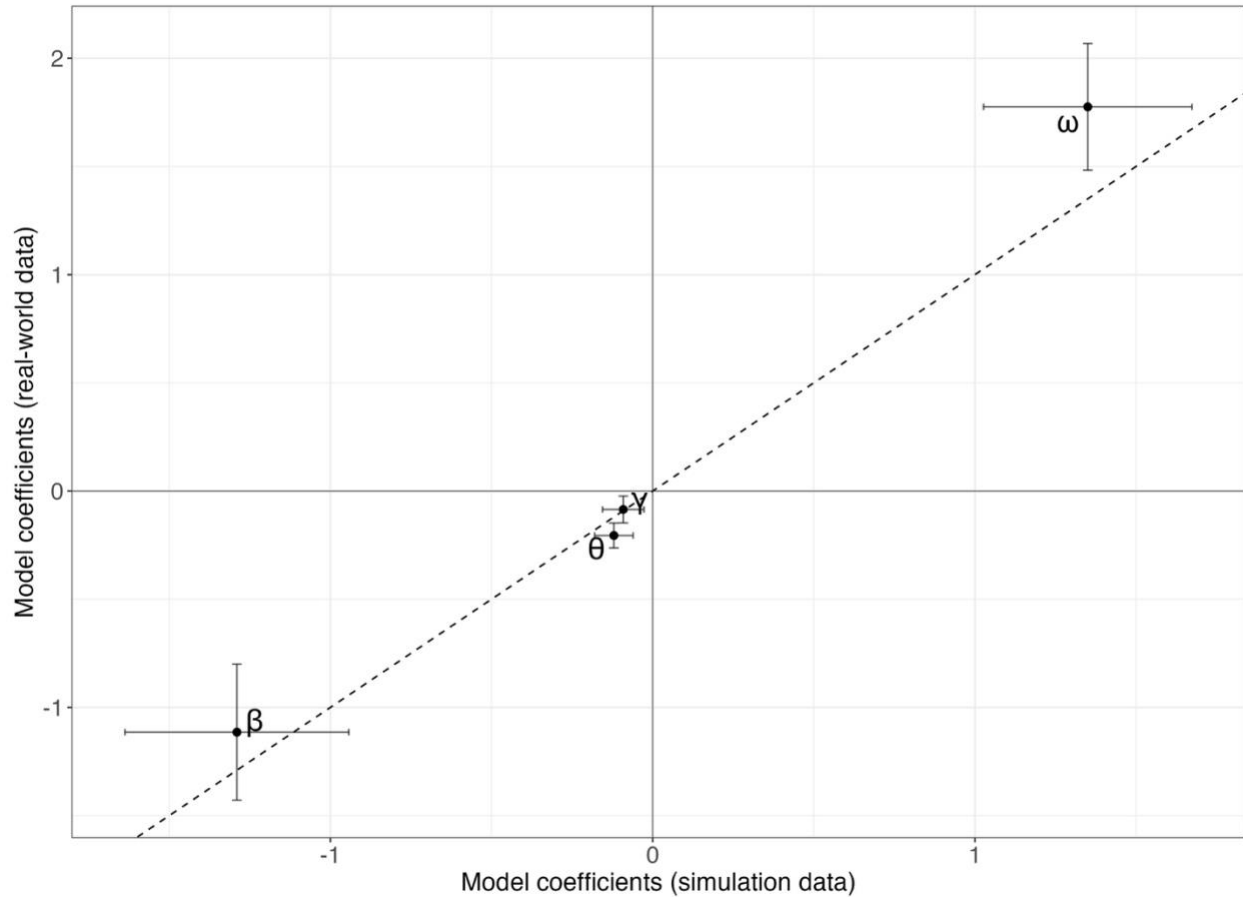
9 **Table S1.** Simulation scenarios and their parameters

Simulation	rho	sigma	alpha	No of iterations
1	25	3	4	20
2	50	3	4	20
3	75	3	4	20
4	100	3	4	20
5	200	3	4	20
6	300	3	4	20
7	400	3	4	20
8	500	3	4	20
9	800	3	4	20
10	1000	3	4	20

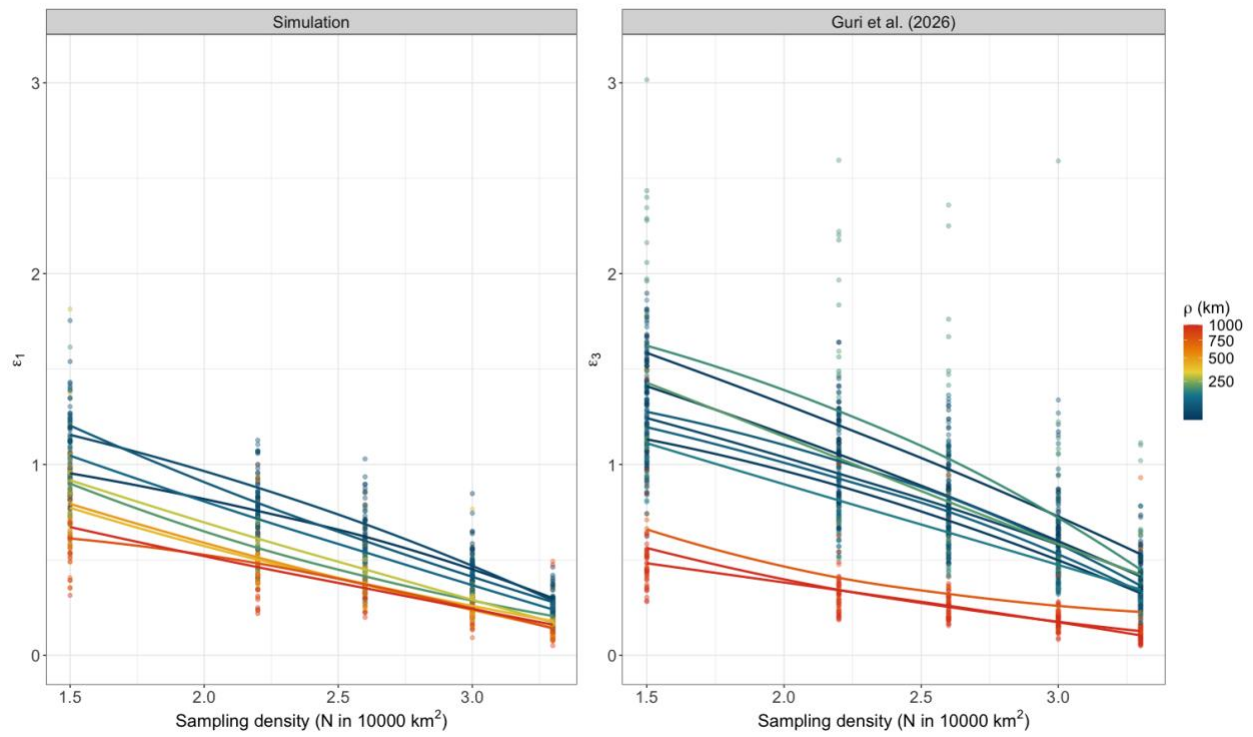
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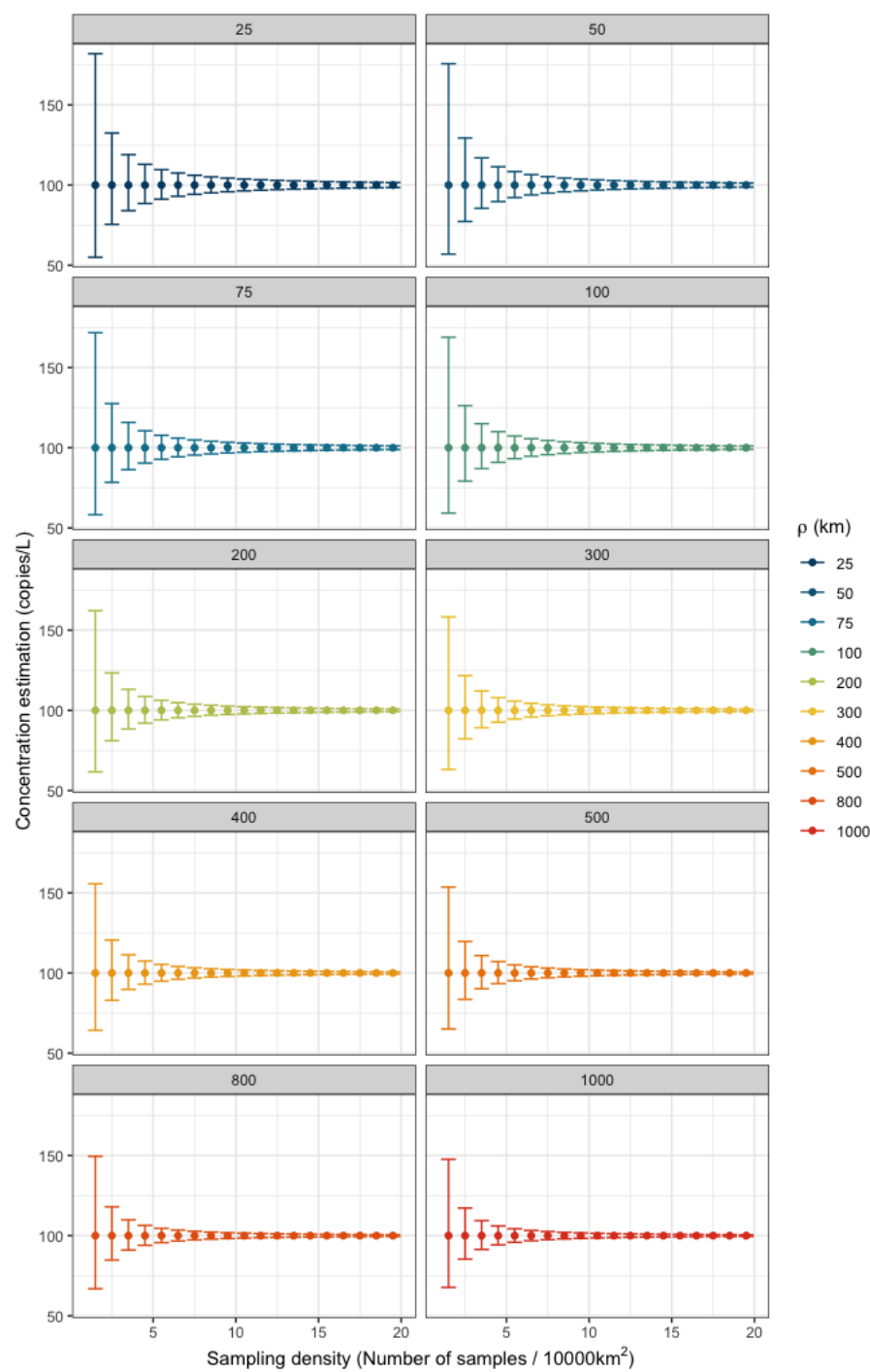
Figure S1. Comparison of regression coefficients estimated from simulated (in-silico in x-axis) datasets and the empirical (in-situ in y-axis) dataset for the model relating prediction error (ϵ) to sampling density (e) and spatial autocorrelation (ρ). Points represent the estimated coefficients for the intercept (ω), sampling density effect (β), spatial autocorrelation effect (θ), and their interaction (γ). Horizontal and vertical error bars indicate ± 2 standard errors of the coefficient estimate from the linear models. The dashed line represents the 1:1 relationship, indicating perfect agreement between simulated and empirical estimates.



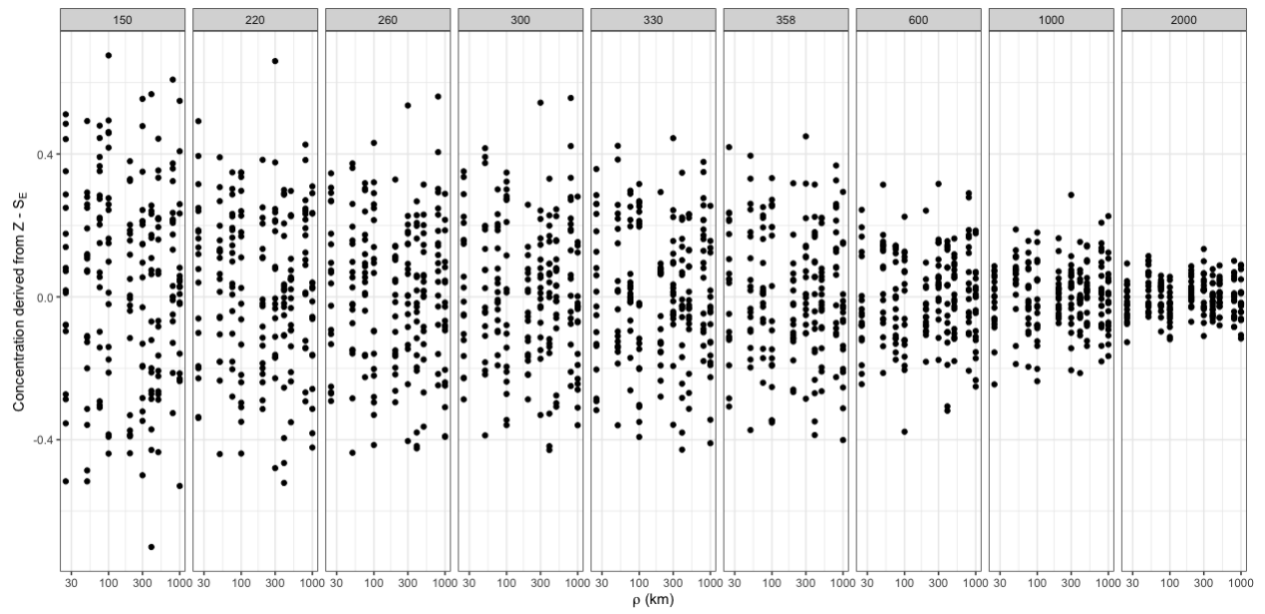
22 **Figure S2.** Comparison of prediction error between the simulated (left) and empirical (right) data
 23 where the baseline Z comes from a dataset of $N = 358$ samples.



25 **Figure S3.** Visualization of the lower and upper bars of prediction given the prediction error (ε)
 26 per sampling effort (E) when the true eDNA concentration is 100 copies/L.



29 **Figure S4.** Bias vs precision, to prove that RMSE used here is mainly accounting for precision
30 and not bias.



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33 **Figure S5.** Showing the observations of the prediction error (dots) and the modeled estimates
 34 (line) from the fitted model using Eq. 6.

